Adaptive Contourlet-Guided Fuzzy Fusion Network for Joint Classification of Hyperspectral and LiDAR Data

Zhipeng Zhong, Cuiping Shi, *Member, IEEE*, Weiwei Sun, *Senior Member, IEEE*, Keyan Wang, *Member, IEEE*, and Yuan Zhang

***Abstract*—In recent years, the joint classification research of hyperspectral images (HSIs) and light detection and ranging (LiDAR) data has garnered significant attention in the field of Earth observation. The information complementarity between them has significantly improved the accuracy of image classification, and multisource remote sensing data fusion has become a current research hotspot. However, existing methods still face some challenges. On the one hand, the ability to capture the fine and smooth contour information is insufficient, which leads to the insufficient extraction of spatial structure features; on the other hand, the lack of modeling the uncertain relationship between modal features restricts the efficient complementarity of cross-modal information. To alleviate the above issues, this article proposes an adaptive contourlet-guided fuzzy fusion network (ACFFN) for joint classification of hyperspectral and LiDAR data. First, an adaptive spectral–spatial contourlet (ASSC) module is proposed, which utilizes LiDAR geometric priors to extract multiscale and multidirectional spatial features, and couples spectral attention to achieve collaborative fusion of spatial–spectral features. Then, a cross-modal feature fusion mechanism based on fuzzy learning is proposed, introducing fuzzy theory to model the uncertainty distribution of features to achieve adaptive cross-modal fusion. In addition, a morphology-guided attention (MGA) module is designed, which enhances elevation structural features through learnable morphological kernels and dynamic attention weights. Experiments and analysis on four public datasets show that our proposed method outperforms some of the most advanced methods in classification performance. The code link of the proposed method is https://github.com/Zzp-12/ACFFN**

***Index Terms*—Contourlet transform, fuzzy logic, hyperspectral image (HSI), light detection and ranging (LiDAR), multimodal.**

I. Introduction

With the rapid development of Earth observation technology, modern spaceborne sensors can obtain massive remote sensing images with submeter resolution, providing important data support for key fields, such as wetland monitoring, urban planning, resource exploration, and precision agriculture [1], [2], [3]. Among them, hyperspectral images (HSIs) can accurately capture the fine reflection characteristics of ground objects by virtue of the hundreds of continuous spectral bands. HSI classification (HSIC) aims to assign category labels to each pixel and has made significant progress with the development of deep learning [4], [5], [6], [7]. However, the spatial resolution of HSI data is relatively limited, making it difficult to fully characterize the geometric structure and spatial details of ground objects; meanwhile, its high sensitivity to lighting conditions makes it prone to spectral distortion in areas such as shadows. These limitations highlight the necessity of multimodal fusion. Light detection and ranging (LiDAR) data, which achieves precise measurement of the 3-D coordinates and reflectance intensity of ground objects by actively emitting laser pulses, can provide elevation information and fine geometric structure features that are lacking in HSI. Therefore, joint classification methods that fuse HSI and LiDAR data have gradually become a research hotspot [8], [9], [10].

In recent years, the breakthrough advancements in deep learning have propelled technological innovations in HSI and LiDAR data classification methods. Unlike traditional approaches that rely on manual feature design, deep learning models can effectively model the complex nonlinear correlations between multimodal data [11], [12], [13]. Among them, convolutional neural networks (CNNs) have become a potent tool for multimodal feature learning with their robust local perception capabilities [14], [15]. Zhang et al. [16] introduced an interleaving perception network that extracts heterogeneous data features through a dual-branch CNN

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architecture. Hang et al. [17] proposed a coupled CNN that achieves efficient joint feature learning through a parameter-shared convolutional structure. Gao et al. [18] combined involution operators and multilayer CNNs to adaptively enhance spatial–spectral features. However, the inherent local receptive-field characteristic of CNNs limits their ability to model long-range dependencies, making it difficult to fully capture global contextual information. Therefore, a series of attention mechanisms have been introduced into joint classification tasks [19], [20]. Zhang et al. [21] developed a mutual guidance attention module to highlight discriminative features by enhancing the information flow between spatial and elevation branches. Zhu et al. [22] proposed a classification method based on multihead axial attention to enhance the model's ability to model multiscale features. Cai et al. [23] introduced a graph attention mechanism to effectively capture long-range dependencies in multimodal data.

On the basis of fully mining single-modal features, the effective fusion of complementary information between heterogeneous modes has become the key to further improving the classification performance. Tang et al. [24] proposed a multiple information collaborative fusion network to explore modal complementarity through the semantic feature space. Xia et al. [25] proposed a comparative learning framework to achieve deep semantic alignment between different modes. Yang et al. [26] proposed a text-supervised multidimensional contrastive network, utilizing text semantic information to guide the complementary fusion of visual features. Guo et al. [27] developed a two-level fusion module, combining cross-attention and low-frequency residuals to address the distribution bias of multisource features in coastal wetlands. Lin et al. [28] proposed a dynamic cross-modal feature interaction network, introducing a dynamic routing mechanism to achieve adaptive feature interaction. Wang et al. [29] adopted a multilevel information fusion strategy, optimizing the stability of classification results through decision-level fusion.

Although the aforementioned methods have demonstrated good performance in joint classification tasks, they still exhibit several limitations.

1) The existing methods lack the ability to capture fine and smooth ground-object contour information. CNN relies on local convolution kernels and pooling operations, which can easily blur pixel-level localization information when extracting depth features, making it difficult to capture large-scale continuous contours. Although Transformer can model long-range dependencies through the attention mechanism, the method of converting images into sequences disrupts the inherent spatial geometric relationships of pixels, resulting in a limited ability to represent land boundaries.

2) The modeling of inherent uncertainty relationships between modal features is insufficient, making it difficult to achieve effective complementarity of cross-modal information.

To alleviate these issues, this article proposes an adaptive contourlet-guided fuzzy fusion network (ACFFN) for joint classification of hyperspectral and LiDAR data. First, an adaptive spectral–spatial contourlet (ASSC) module is proposed, which extracts spatial features with the aid of geometric prior information from LiDAR, and couples spectral attention to achieve deep collaborative fusion of spatial–spectral features. Then, a cross-modal feature fusion mechanism based on fuzzy learning is proposed, introducing fuzzy theory to model feature uncertainty distribution to achieve adaptive cross-modal information aggregation. In addition, a morphological-guided attention mechanism is carefully designed to achieve targeted enhancement of elevation structural features. Extensive experiments conducted on four public datasets demonstrate that our proposed method outperforms some current mainstream advanced methods in terms of classification performance.

The main contributions of our work are as follows.

1) To accurately capture the inherent complex geometric features in HSI, an ASSC module is proposed. This module utilizes the gradient information of LiDAR data as geometric structure priors, models multiscale and multidirectional spatial features through an adaptive nonsubsampled contourlet transform, and designs a dynamic coordination mechanism to achieve the collaborative fusion of spectral–spatial features.

2) A cross-modal feature fusion mechanism based on fuzzy learning block (FLB) is constructed. By introducing the fuzzy system theory to model the uncertainty distribution of multimodal features, it dynamically captures the complementary and common relationships between features, and achieves adaptive fusion.

3) A morphology-guided attention (MGA) module is designed, which enhances the structural sensitivity and morphological preservation ability of elevation information through learnable morphological kernels and dynamic weight allocation of attention.

II. Related Work

A. Contourlet Transform

The contourlet transform [30] is a core method of multiscale geometric analysis. It achieves efficient sparse representation of curve singularities in images through a collaborative architecture of directional filter banks and pyramid decomposition, significantly enhancing the representation ability of 2-D geometric features. In recent years, scholars have attempted to combine the contourlet transform with deep learning to construct feature enhancement frameworks guided by geometric priors [31], [32]. Liu et al. [33] proposed a contourlet CNN that integrates subband features into a convolutional architecture to enhance the network's ability to learn texture features. Zhang et al. [34] proposed an unsupervised method that alleviates the problem of insufficient geometric representation by introducing a nonsubsampled contourlet. Liu et al. [35] combined the contourlet transform with Mamba UNet to achieve high-fidelity fusion of infrared and visible light images. The successful practice of these works reflects the excellent performance of the contourlet transform in geometric representation. This article is the first to extend this technology to the classification research of HSI and LiDAR, filling the technological gap in this field.

B. Fuzzy System

The fuzzy theory was proposed by Zadeh [36]. Its core is to quantitatively describe the membership relationships of sets

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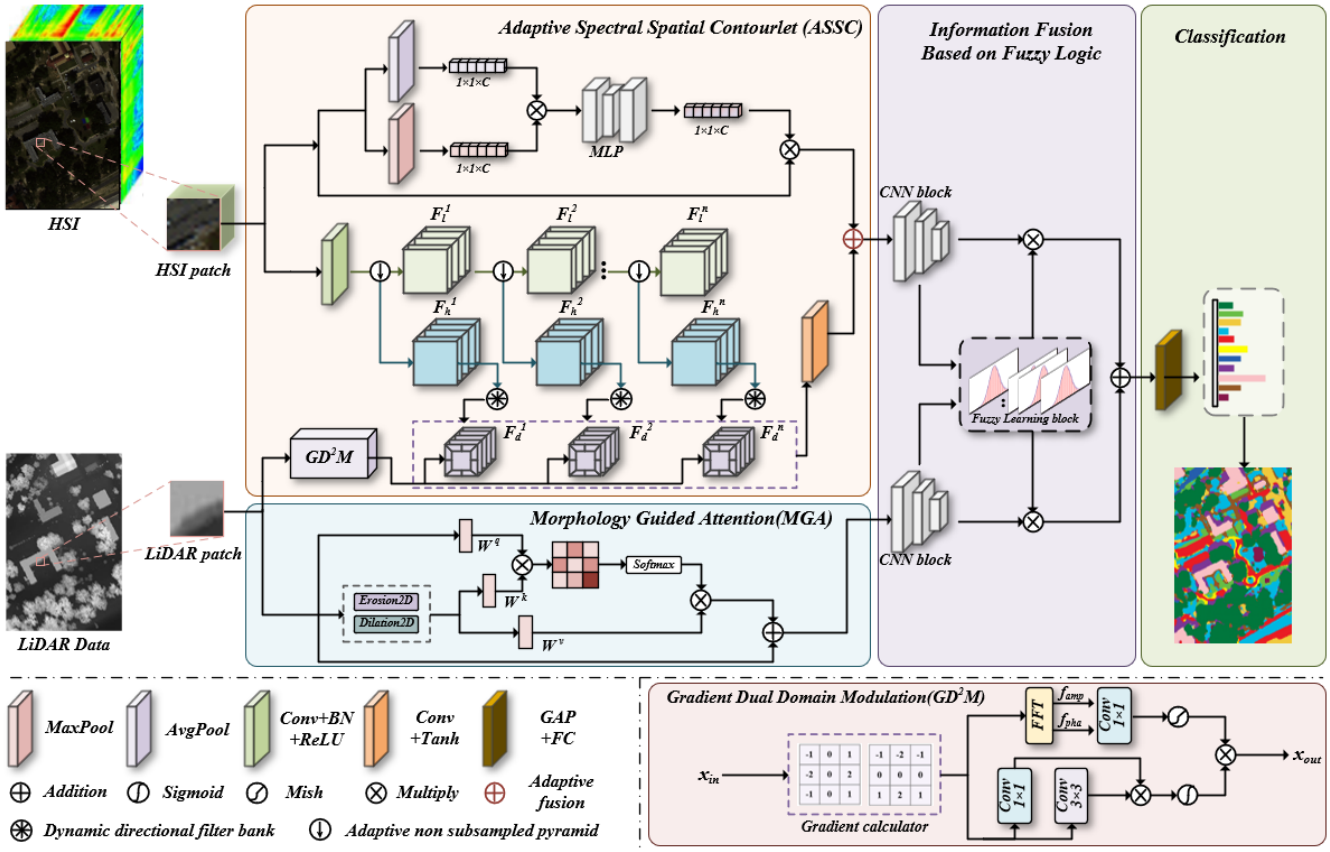


Fig. 1. Schematic of the proposed ACFFN.

through membership functions, breaking through the limitations of classical binary logic and providing theoretical support for dealing with uncertainty and ambiguity. The robustness of the deep learning model to uncertain factors, such as fuzzy boundary, noise interference, and modal difference, is weak. The combination of fuzzy theory and deep learning has become an important way to solve the above problems and has made significant progress, especially in the field of medical image analysis [37], [38]. Ding et al. [39] proposed a multigranularity depth neural network, designed membership functions to calculate fuzzy features, and effectively guided histopathological feature learning. Lin et al. [40] proposed a cross-scale fuzzy holistic attention network, which integrates the fuzzy theory into the deep attention module to realize the adaptive focusing of the retinopathy area and the suppression of irrelevant information. However, the application of fuzzy theory in remote sensing data analysis has not received sufficient attention, and the task of multisource remote sensing data fusion itself generally has inherent uncertainty. Specifically, the spectral information reliability of HSI in shaded or cloud-covered areas will significantly decrease, while the penetration ability of LiDAR data in densely vegetated areas or complex building structures is inherently limited. In addition, the boundary of land features has transitional and mixed pixel problems in space, and traditional weighted fusion is difficult to effectively handle such inherent fuzzy semantic boundaries. This research innovatively introduces the fuzzy theory into the joint classification task of HSI and LiDAR, and proposes a fusion mechanism based on FLB, which effectively reduces the uncertainty of fused multimodal features, realizes the complementary fusion of cross-modal information, and provides a new theoretical perspective for intelligent perception in complex remote sensing scenes.

III. Methodology

The proposed ACFFN is illustrated in Fig. 1. The HSI patch enters the ASSC module for the extraction of spatial–spectral features, where the contourlet transform is introduced to capture spatial multiscale and multidirectional features, and the dynamic collaborative mechanism is utilized to achieve the fusion of spatial–spectral features. Meanwhile, the LiDAR patch is processed in two ways: one is input into the ASSC module, using its elevation gradient as a geometric prior to guide the precise modeling of HSI edge features; the other is input into the MGA module, where learnable morphological kernels are combined to enhance the elevation information. Subsequently, the HSI spatial–spectral features extracted by ASSC and the LiDAR elevation features enhanced by MGA are input into a cross-modal fusion mechanism based on FLB for adaptive deep fusion. Finally, the fused features are sent to the classification head to obtain the classification results.

A. Adaptive Spectral–Spatial Contourlet

Existing methods still have limitations in the fine characterization of complex terrain boundaries, making it difficult

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to accurately capture the continuity and integrity of contours. Compared to the traditional contourlet transform, the nonsubsampled contourlet transform solves the problems of spectral aliasing and translation sensitivity while retaining the advantage of anisotropy, significantly enhancing the modeling ability of contour features. To accurately model the spatial geometric features of HSI while maintaining its spectral complexity, this article designs an ASSC module based on the nonsubsampled contourlet transform. Specifically, the HSI patch $X_{\text{hsi}} \in \mathbb{R}^{H\times W\times C}$ is initially processed using a convolution block, which can be represented as follows:

$$F_{hsi} = ReLU\left(BN\left(Conv_{1\times 1}\left(X_{\text{hsi}}\right)\right)\right) \tag{1}$$

where $Conv_{k\times k}(\cdot)$, $BN(\cdot)$, and $ReLU(\cdot)$ represent convolution with kernel size k × k, batch normalization, and activation function ReLU, respectively. Then, the HSI features are decomposed into low-frequency and high-frequency subbands through an adaptive nonsubsampled pyramid (ANSP). This process can be expressed as

$$F_l, F_h = f_{ANSP}(F) \tag{2}$$

where $f_{ANSP}(\cdot)$ represents ANSP, which decomposes the input feature F into low-frequency subbands F_l and high-frequency subbands F_h. Specifically, low-pass filtering is used to extract low-frequency subbands; the residual of the original features and low-frequency components is adaptively modified through high-frequency compensation filtering to obtain high-frequency subbands. This process can be expressed as

$$\begin{cases} F_l^{i+1} = \mathcal{G} * F^i \\ F_h^{i+1} = F^i - F_l^{i+1} + \mathcal{H} * F^i \end{cases} \tag{3}$$

where $*$ represents the convolution operation, $\mathcal{G}$ and $\mathcal{H}$ represent the Gaussian low-pass filter operator and the high-frequency compensation filter operator, respectively. Apply the designed dynamic directional filter bank (DDFB) to the high-frequency subbands F_h to extract its multidirectional texture components, i.e., directional subbands F_d. As shown in the following formula

$$F_d = f_{DFB}(F_h, X_{lidar}) \tag{4}$$

where $f_{DFB}(\cdot)$ is DDFB. First, dynamic modulation is performed on the high-frequency subbands of HSI using LiDAR data to achieve preliminary alignment of spectral geometric features. Subsequently, a directional filter bank is constructed to extract feature responses in various directions. Furthermore, the gradient information of LiDAR data is used as a geometric prior to generate a geometric attention map through gradient dual-domain modulation (GD2M). Then, it is combined with directional features to enhance the directional representation ability of ground-object boundaries. This specific process is outlined in Algorithm 1. GD2M first uses the Sobel operator to calculate the spatial gradient of LiDAR data, and maps the features to the frequency domain. Based on this, terrain prior guidance is provided for multidirectional feature fusion through joint modeling in both spatial and frequency domains. See Algorithm 2 for the specific process.

The decomposition-filtering process is recursively executed in the low-frequency subbands: features are decomposed into high-frequency and low-frequency components through ANSP, and high-frequency components are subject to DDFB processing, which can build multiscale and multidirectional feature representation. The specific decomposition level is set to 3. Finally, all directional subbands are spliced and reconstructed to obtain the spatial feature F_{spa}. The process can be represented as

$$F_{spa} = Tanh\left(Conv_{1\times 1}\left(Concat\left(F_d^1, F_d^2, \ldots, F_d^n\right)\right)\right). \tag{5}$$

Among them, $Concat(\cdot)$ and $Tanh(\cdot)$ represent the concatenation and activation functions Tanh, respectively. Considering the importance of spectral features in feature discrimination, a spectral branch is designed to enhance the discriminative band response and optimize the feature representation. The steps are as follows:

$$\begin{cases} g_{avg} = Avgpool\left(X_{\text{hsi}}\right) \\ g_{\max} = Maxpool\left(X_{\text{hsi}}\right). \end{cases} \tag{6}$$

Among them, $Avgpool(\cdot)$ and $Maxpool(\cdot)$ represent average pooling and maximum pooling, respectively. To fully retain the global and significant spectral information, the two kinds of pooled outputs are multiplied at the element level, and the fusion results are input into multilayer perceptrons (MLPs) to generate spectral attention weights. Finally, the weight is multiplied by the original input features to enhance the

Algorithm 1 Dynamic Directional Filter Bank

Input : High frequency subband $\boldsymbol{F}_{\text{h}} \in \mathbb{R}^{H\times W\times C}$
LiDAR data $\boldsymbol{X}_{\text{lidar}} \in \mathbb{R}^{H\times W\times 1}$
Output: Directional subband $\boldsymbol{F}_{\text{d}} \in \mathbb{R}^{H\times W\times C}$
Initialize base kernel $\boldsymbol{W}_{\text{base}} \in \mathbb{R}^{k\times k\times C}$; // Filter kernel
begin
  // LiDAR-guided feature modulation
  $\boldsymbol{P} \leftarrow \texttt{Project}(\boldsymbol{X}_{\text{lidar}})$; // Conv3×3→ReLU→Conv1×1
  $\boldsymbol{F}'_{\text{h}} \leftarrow \boldsymbol{F}_{\text{h}} \odot \texttt{Sigmoid}(\boldsymbol{P})$
  // Directional Filter Bank
  foreach *direction* $i \leftarrow 0$ **to** 7 **do**
    $\boldsymbol{W}_i \leftarrow \texttt{Rotate}(\boldsymbol{W}_{\text{base}}, i)$; // Generate rotated kernel
    $\boldsymbol{F}_i \leftarrow \boldsymbol{W}_i * \boldsymbol{F}'_{\text{h}}$; // Directional convolution
  // Geometric prior
  $\boldsymbol{A} \leftarrow GD^2M(\boldsymbol{X}_{\text{lidar}})$; // Gradient Dual Domain Modulation
  // Weighted fusion
  $\boldsymbol{F}_{\text{weighted}} \leftarrow \sum_{i=0}^{7} \boldsymbol{A}_i \odot \boldsymbol{F}_i$
  $\boldsymbol{F}_{\text{d}} \leftarrow \boldsymbol{F}'_{\text{h}} + \boldsymbol{F}_{\text{weighted}}$
  return $\boldsymbol{F}_{\text{d}}$

Algorithm 2 Gradient Dual-Domain Modulation

Input : LiDAR data $X_{lidar} \in R^{H\times W\times 1}$
Output: Geometric attention $A \in R^{H\times W\times D}$
begin
  // Gradient extraction
  $\nabla X \leftarrow W_{sobel} * X_{lidar}$; // Sobel operators
  // Frequency domain transformation
  $\mathcal{F} \leftarrow \texttt{FFT}(\nabla X)$
  $M \leftarrow \log(1 + |\mathcal{F}|)$; // Magnitude spectrum
  $\Phi \leftarrow \angle\mathcal{F}$; // Phase spectrum
  // Frequency branch
  $F_{freq} \leftarrow GELU\left(\texttt{Conv}_{1\times 1}(M \odot \Phi)\right)$
  // Spatial branch
  $F_{sp} \leftarrow Mish\left(\texttt{Conv}_{3\times 3}(\nabla X)\right) \odot Mish\left(\texttt{Conv}_{1\times 1}(\nabla X)\right)$
  // Cross-domain modulation
  $A \leftarrow \texttt{Sigmoid}(F_{sp}) \odot F_{freq}$
  return A

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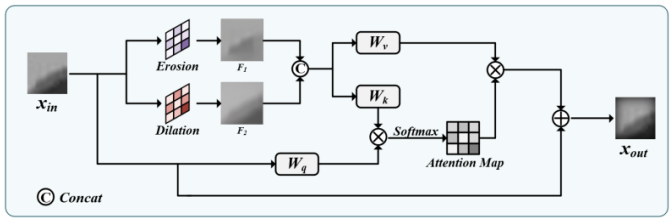


Fig. 2. Schematic of MGA.

discrimination ability of spectral features. It can be expressed as

$$F_{spe} = BN\left(X_{\text{hsi}} \odot MLP\left(g_{avg} \odot g_{\max}\right)\right) \tag{7}$$

where $MLP(\cdot)$ and $\odot$ represent MLP and element multiplication, respectively. Finally, the spectral and spatial features are weighted and fused to obtain the feature F_{ss}. Specifically, spectral and spatial features are first connected along the channel dimension, cross-modal correlation is learned using the convolutional structure, and the channel is mapped to 2, corresponding to the weight of the two features. Finally, residual connection is used to alleviate the disappearance of the gradient. This process can be expressed as

$$G = Softmax\left(Conv_{1\times1}\left(ReLU\left(Conv_{1\times1}\left(Concat\left(F_{spe}, F_{spa}\right)\right)\right)\right)\right) \tag{8}$$

$$F_{ss} = G[0] \odot F_{spe} + G[1] \odot F_{spa} + X_{\text{hsi}} \tag{9}$$

where $Softmax(\cdot)$ represents the activation function Softmax, which is used to normalize the weight vector.

B. Morphology-Guided Attention

The existing methods have deficiencies in modeling the fine-grained features of LiDAR elevation data, and traditional attention mechanisms often result in the loss of terrain details due to their global aggregation characteristics. To this end, this article proposes a targeted MGA module, as shown in Fig. 2. The specific process is as follows:

$$F_{di} = \sigma\left(W_{di}\right) {}^{*}X_{lidar} \tag{10}$$

$$F_{er} = \left(1 - \sigma\left(W_{er}\right)\right) {}^{*}X_{lidar} \tag{11}$$

$$F_{m} = Concat\left(F_{di}, F_{er}\right) \tag{12}$$

where $\sigma(\cdot)$ represents the Sigmoid function, and W_{di} and W_{er} represent the learnable dilation kernel and erosion kernel, respectively. The results of dilation and erosion operations are connected along the channel dimension to obtain a morphological feature F_m. Then, the original feature is used to construct the query vector Q, and the morphological feature is used to construct the key vector K and the value vector V. In addition, the element multiplication is used to replace the matrix multiplication to efficiently construct the attention weight map A. Finally, the skip connection is used to prevent the network from overfitting. The specific process is as follows:

$$\begin{cases} Q = W_Q * X_{lidar} \\ K = W_K * F_m \\ V = W_V * F_m \end{cases} \tag{13}$$

$$A = Softmax\left(\frac{Q \odot K}{\sqrt{d}}\right) \tag{14}$$

$$F_{el} = A \odot V + X_{lidar} \tag{15}$$

where F_{el} represents the features of the final MGA output, and d is the spatial dimension of Q.

C. Feature Fusion Based on Fuzzy Logic

The high-dimensional continuous distribution of HSI spectral features and the spatial-discrete gradient of LiDAR elevation features are essentially heterogeneous. Traditional cross-modal fusion methods are difficult to achieve effective information fusion. For this reason, we propose a fusion mechanism based on FLB to embed the uncertainty modeling depth into the fusion process: quantify the correlation between modes through the membership function in fuzzy logic, adaptively guide spatial–spectral features and elevation features to achieve complementary advantages in the fusion, and achieve the depth collaboration of cross-modal information

$$F'_{ss} = ConvStack^{(3)}_{c\to128}\left(F_{ss}\right) \tag{16}$$

$$F'_{el} = ConvStack^{(3)}_{1\to128}\left(F_{el}\right) \tag{17}$$

where $ConvStack^{(k)}_{a\to b}(\cdot)$ represents the convolution block, superscript represents the number of convolution layers (where k = 3), and subscript represents the change of channel number. This module is used to unify the channel dimensions of cross-modal features and also serves as the baseline network. Then, the processed features are input into the FLB module for fuzzy modeling

$$A_{ss} = \sigma\left(FLB\left(F'_{ss}\right)\right) \tag{18}$$

$$A_{el} = \sigma\left(FLB\left(F'_{el}\right)\right) \tag{19}$$

where $FLB(\cdot)$ represents FLB. As shown in Fig. 3, FLB first uses the Gaussian membership function to calculate the membership degree of the feature to obtain the membership tensor $Z_{m,c,h,w}$, and then uses fuzzy logic to aggregate the membership tensor: the AND operation is realized through geometric averaging to capture global dependencies, and the OR operation is realized through maximum value operation to extract local salient features. The specific process is as follows:

$$Z_{m,c,h,w} = \exp\left(-\left(\frac{X_{c,h,w} - \mu_{m,c}}{\sigma_{m,c}}\right)^2\right) \tag{20}$$

$$Z_{and} = \exp\left(\frac{1}{M}\sum_{m=1}^{M}\log\left(Z_{m,c,h,w} + \varepsilon\right)\right) \tag{21}$$

$$Z_{or} = \max_m Z_{m,c,h,w}. \tag{22}$$

In which, ε represents the extremely decimal avoiding Log (0). To sharpen the distinction between features, the fixed temperature coefficient τ (set to 0.05) is introduced to normalize the two indices, respectively. Then, the normalized global dependent features ${Z_{and}}^{norm}$ and local salient features ${Z_{or}}^{norm}$ are connected along the channel dimension and fed into the convolution layer for nonlinear fusion to obtain the feature representation Y enhanced by fuzzy logic. This process is as follows:

$${Z_{and}}^{norm} = \frac{\exp\left(-Z_{and}/\tau\right)}{\sum_c \exp\left(-Z_{and}/\tau\right)} \tag{23}$$

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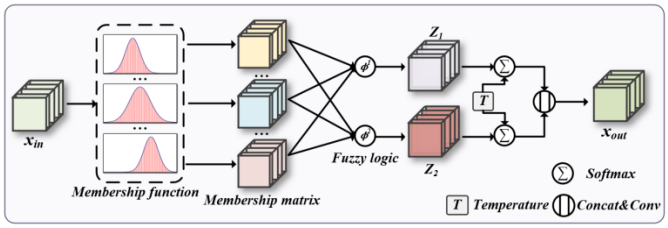


Fig. 3. Schematic of FLB.

$$Z_{or}{}^{norm} = \frac{\exp\left(-Z_{or}/\tau\right)}{\sum_c \exp\left(-Z_{or}/\tau\right)} \tag{24}$$

$$Y = ReLU\left(Conv_{3\times3}\left(Concat\left(Z_{and}{}^{norm}, Z_{or}{}^{norm}\right)\right)\right). \tag{25}$$

Then, the preprocessed features are cross-modality complementarily fused using the correlation weights generated by FLB to obtain the final feature representation F_{fusion}

$$F_{fusion} = F'_{ss} \odot A_{ss} + F'_{el} \odot A_{el}. \tag{26}$$

Finally, F_{fusion} is sent to the classification head, composed of global average pooling and a linear layer for category prediction.

IV. Experimental Results and Analysis

A. Dataset Description

Houston2013 Dataset: This dataset was collected from the University of Houston and surrounding urban areas and includes 144-band HSI (0.38–1.05 μm) and single-band LiDAR data. The image size is 349 × 1905 pixels, and the spatial resolution is 2.5 m. It provides 15 categories and a total of 15 029 real samples.

Trento Dataset: This dataset is acquired from a rural area in the south of Trento, Italy. Among them, HSI contains 63 bands with a spectral coverage of 0.42–0.99 μm; LiDAR data is a single band. The spatial size of the dataset is 166 × 600, and the resolution is 1 m. It includes six different feature categories and provides 30 214 real samples for field verification.

MUUFL Dataset: This dataset is collected in the campus area of Gulf Park University in Southern Mississippi. Its HSI contains 72 bands, with a spectral coverage of 0.38–1.05 μm; LiDAR data is dual-band (wavelength 1.06 μm). The spatial size of the dataset is 325 × 220 pixels, including 11 different ground feature categories, and a total of 53 687 real samples.

SZU Dataset: This dataset was collected at the Yuehai Campus of Shenzhen University [41]. Among them, HSI contains 98 bands with a spectral coverage of 0.399–0.924 μm; LiDAR data is a single band. The spatial size of the dataset is processed to 771 × 601 pixels, including 20 different tree species categories, with a total of 98 347 real samples. Fig. 4 shows the pseudocolor image of HSI and the grayscale image of LiDAR for each dataset. The detailed category information and the number of samples of each dataset are shown in Table I.

B. Experimental Setup

All experiments, including the proposed method and comparative deep learning approaches, were conducted using PyTorch. The hardware configuration comprised an Intel i5-12400F CPU, NVIDIA RTX 4060Ti GPU, and 32 GB RAM. Network optimization employed the Adam optimizer with an initial learning rate of 1e-4, patch size set to 10, batch size set to 64, and the epoch is set to 500.

We used four common evaluation metrics: overall accuracy (OA), average accuracy (AA), kappa coefficient ($\kappa \times 100$), and each category accuracy. Higher values correspond to better performance.

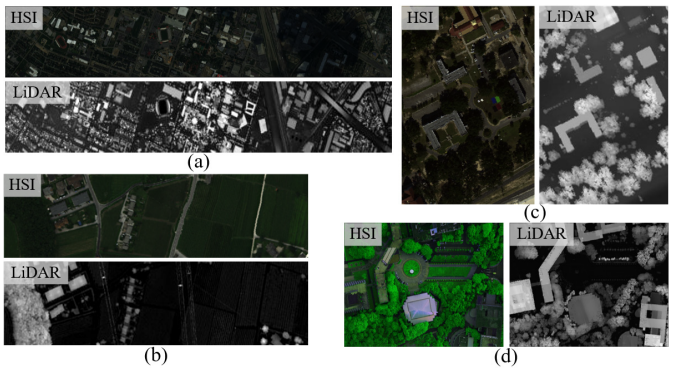


Fig. 4. Pseudocolor image and LiDAR grayscale image. (a) Houston 2013. (b) Trento. (c) MUUFL. (d) SZU.

C. Ablation Study

ASSC, MGA, and FLB are the core modules of the proposed method. To verify the effectiveness of these modules, we conducted ablation experiments, and the results are shown in Table II. In case 1, the input HSI patch and LiDAR patch are connected after the baseline convolution block processing, and directly enter the classification head for category prediction. Owing to simple convolution processing and the direct connection without deep fusion, the classification performance is poor. Case 2 adds an ASSC module on this basis to perform multidirectional contourlet decomposition and reorganization of HSI features. The OA of the four datasets increased significantly, especially the Houston 2013 dataset, with an increase of 1.03%. Case 3 introduces an MGA module on the basis of case 1, improves the structural sensitivity of LiDAR elevation information by introducing morphological features, and further effectively improves the OA index. Case 4 integrates ASSC and MGA, and the synergy of the two makes the classification performance better. Case 5 is the complete structure of the network. On the basis of Case 4, an FLB module is added to achieve cross-modal feature fusion. Further improved OA verifies the effectiveness of introducing uncertainty modeling into cross-modal joint classification. Compared with the baseline network, the OA of the four datasets increases by 1.33%, 0.32%, 1.57%, and 1.92%, respectively. In short, the classification performance of each module has been significantly improved, and each module can work effectively.

D. Performance Comparison

To verify the effectiveness of the proposed ACFFN model, we compared it with several of the most advanced HSI and

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR FOUR DATASETS

NO.	Houston 2013			Trento			MUUFL			SZU		
	Class	Training	Test	Class	Training	Test	Class	Training	Test	Class	Training	Test
1	Healthy Grass	20	1231	Apple Trees	20	4014	Trees	20	23226	Terminalia arjuna	100	6856
2	Stressed Grass	20	1234	Buildings	20	2883	Mostly Grass	20	4250	Kigelia africana	100	1290
3	Synthetic Grass	20	677	Ground	20	459	Mixed Ground Surface	20	6862	Roystonea regia	100	1409
4	Trees	20	1224	Woods	20	9103	Dirt and Sand	20	1806	Ficus macrophylla	100	25633
5	Soil	20	1222	Vineyard	20	10481	Road	20	6667	Delonix regia	100	1398
6	Water	20	305	Roads	20	3154	Water	20	446	Archontophoenix alexandrae	100	132
7	Residential	20	1248				Buildings Shadow	20	2213	Litchi chinensis	100	8629
8	Commercial	20	1224				Buildings	20	6220	Mangifera indica	100	1359
9	Road	20	1232				Sidewalk	20	1365	Araucaria cunninghamii	100	5447
10	Highway	20	1207				Yellow Curb	20	163	Livistona chinensis	100	775
11	Railway	20	1215				Cloth Panels	20	249	Dracontomelon duperreanum	100	1863
12	Parking Lot 1	20	1213							Acacia auriculiformis	100	5316
13	Parking Lot 2	20	449							Ficus concinna	100	12044
14	Tennis Court	20	408							Ficus elastica	100	10254
15	Running Track	20	640							Leucaena leucocephala	100	238
16										Bauhinia purpurea	100	449
17										Camphora officinarum	100	2214
18										Dimocarpus longan	100	4922
19										Bombax ceiba	100	844
20										Corymbia citriodora	100	5275
-	Total	300	14729	Total	120	30094	Total	220	53467	Total	2000	96347

TABLE II
RESULTS OF ABLATION EXPERIMENTS

Case	Modules			OA (%)			
	ASSC	MGA	FLB	Houston 2013	Trento	MUUFL	SZU
1	-	-	-	94.24	99.15	84.44	90.32
2	√	-	-	95.27	99.26	85.04	91.33
3	-	√	-	94.89	99.29	84.75	90.53
4	√	√	-	95.36	99.37	85.20	92.02
5	√	√	√	95.57	99.47	86.01	92.24

LiDAR joint classification models, including S^2Enet [8], GLT [19], MFT [20], CALC [9], Exvit [42], FDNet [10], CrossHL [43], S3F2Net [29], and MCFNet [44]. All experiments were carried out in the same environment, and the network was set according to the parameters described in the original text.

1) Analysis of Classification Results: Tables III–VI show the OA, AA, $\kappa \times 100$ of all methods on the four datasets, and the accuracy of each class, respectively. The best results are shown in bold; Figs. 5 and 6 further show the classification diagram of each method on some datasets. The results show that the proposed ACFFN method is significantly better than other methods in classification results, and OA is 0.45%, 0.19%, 0.53%, and 1.36% higher than the suboptimal method, respectively. Due to its relatively simple network architecture and cross-modal fusion approach, MFT has limited classification performance. FDNet and MCFNet are combined with the discrete wavelet transform to verify the effectiveness of frequency-domain features in rich spatial feature representation. GLT exhibits outstanding performance through the processing of multiscale spatial features. Exvit and CrossHL introduce cross-modal attention mechanisms, effectively enhancing the interaction of cross-modal features and outperforming most baseline methods in terms of performance. Overall, methods that combine multiscale analysis, such as GLT, MCFNet, and our proposed model, have shown excellent classification performance. On the Houston 2013 dataset, our model achieved the best performance in seven categories, with categories such as railways and highways performing particularly well, highlighting the effectiveness of combining geometric prior information and morphological

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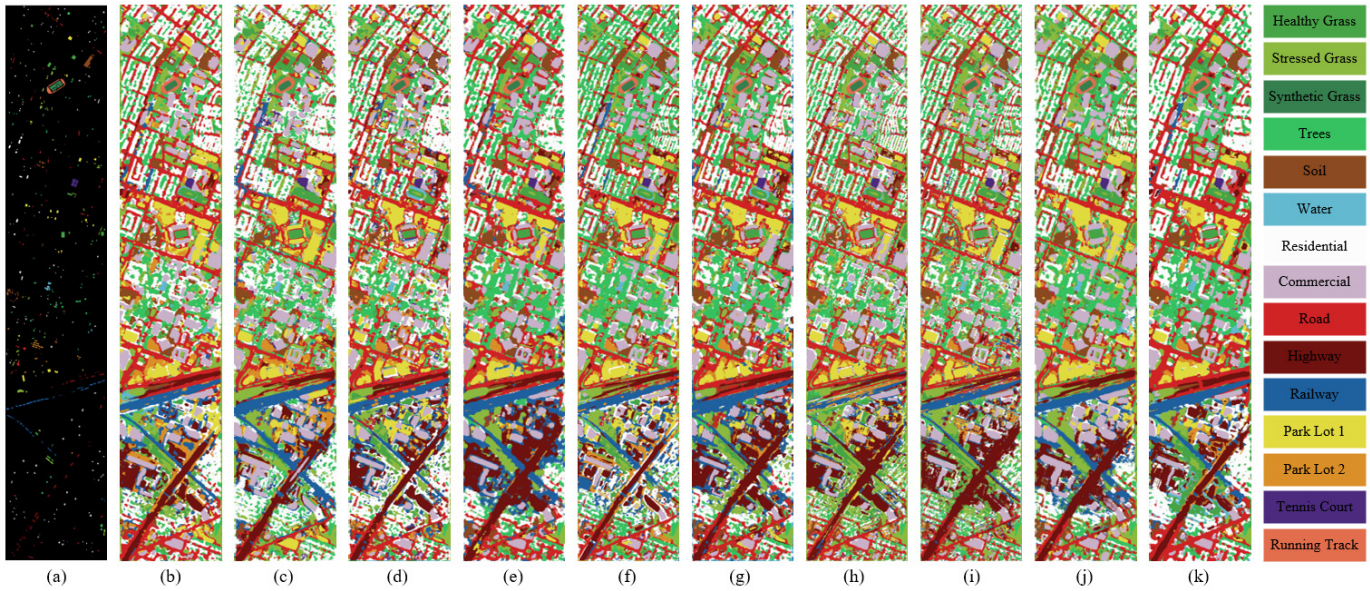


Fig. 5. Classification maps of the Houston 2013 dataset. (a) Ground-truth map. (b) S^2Enet (94.53%). (c) GLT (95.12%). (d) MFT (92.17%). (e) CALC (93.62%). (f) Exvit (93.74%). (g) FDNet (93.56%). (h) CrossHL (93.53%). (i) S3F2Net (94.50%). (j) MCFNet (94.73%). (k) Proposed (95.57%).

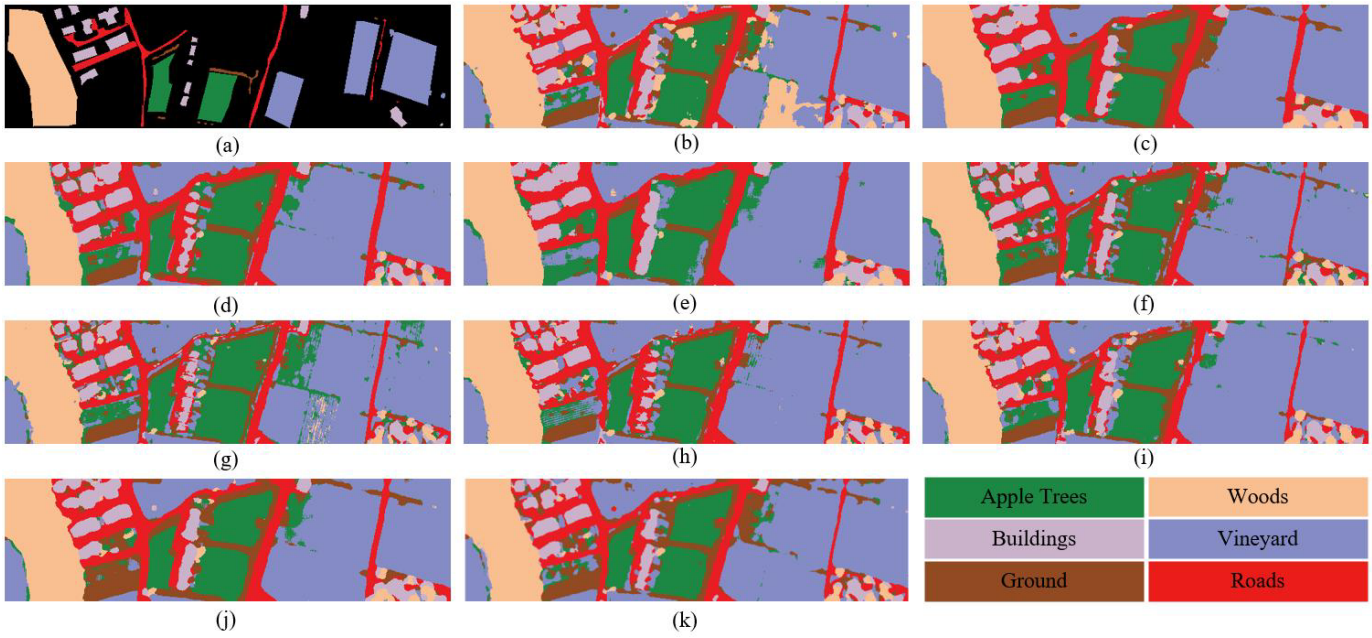


Fig. 6. Classification maps of the Trento dataset. (a) Ground-truth map. (b) S^2Enet (99.14%). (c) GLT (99.28%). (d) MFT (98.53%). (e) CALC (99.17%). (f) Exvit (99.20%). (g) FDNet (98.62%). (h) CrossHL (98.57%). (i) S3F2Net (98.85%). (j) MCFNet (99.22%). (k) Proposed (99.47%).

knowledge to model spatial features for identifying complex categories. Although the classification task of the Trento dataset is relatively easy, the proposed method still maintains a leading position in all evaluation metrics and most categories, demonstrating the robustness of the method. In the MUUFL dataset, our model still achieved the best results, mainly due to the introduction of a cross-modal fusion strategy using fuzzy logic, which effectively solved the modal differences and class boundary blurring problems between the two types of data, thereby achieving more accurate classification results. In addition, experiments on the SZU dataset further verify the generalization of the proposed method. The dataset covers a variety of tree species, with high similarity of spectral features and significant differences in spatial structure, which puts forward higher requirements for the feature discrimination ability of the classification model. The proposed method enhanced the capture of crown shape and edge structure through contourlet transform, and combined with fuzzy fusion mechanism to coordinate the uncertainty of multisource features, thereby achieving the best accuracy in most tree species, especially in the distinction between Ficus concinna and Ficus macrophylla, which showed that the proposed method was suitable for

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TABLE III
CLASSIFICATION RESULTS OF HOUSTON 2013 DATASET

Class	S^2ENet	GLT	MFT	CALC	Exvit	FDNet	CrossHL	S3F3Net	MCFNet	Proposed
1	96.05	92.77	91.96	89.60	93.99	91.47	95.61	96.26	87.81	**96.91**
2	99.03	98.95	98.06	**99.84**	98.87	98.87	97.41	98.95	99.59	99.03
3	99.61	99.11	98.97	**99.85**	99.11	99.70	99.70	**99.85**	99.56	99.70
4	93.71	**95.51**	92.48	93.63	93.71	93.95	94.12	94.59	93.95	93.06
5	**100.00**	99.92	99.51	**100.00**	99.92	99.92	**100.00**	**100.00**	**100.00**	**100.00**
6	99.13	96.07	99.02	98.69	97.70	**100.00**	96.39	99.34	96.07	97.70
7	**98.45**	96.79	97.36	94.39	97.92	95.01	95.27	95.35	96.96	98.16
8	74.13	**83.17**	76.55	80.88	81.29	75.25	80.64	77.21	81.86	81.78
9	84.60	91.15	74.43	82.71	78.90	78.98	87.34	84.09	90.18	**91.80**
10	98.09	99.17	93.12	99.09	96.35	97.85	93.87	97.51	98.51	**99.75**
11	97.12	95.14	94.98	92.51	95.06	96.05	86.83	96.69	96.13	**97.45**
12	94.17	90.68	90.68	92.83	90.35	**96.62**	93.16	94.71	94.23	91.76
13	98.29	**98.66**	96.44	95.77	98.44	94.88	97.10	97.55	96.44	93.99
14	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
15	**100.00**	**100.00**	97.66	99.69	**100.00**	99.53	**100.00**	98.91	**100.00**	**100.00**
OA(%)	94.53	95.12	92.17	93.62	93.74	93.56	93.53	94.50	94.73	**95.57**
AA(%)	95.49	95.81	93.41	94.63	94.77	94.60	94.50	95.40	95.42	**96.07**
$\kappa\times100$	94.09	94.72	91.53	93.10	93.23	93.03	93.00	94.05	94.30	**95.21**

TABLE IV
CLASSIFICATION RESULTS OF TRENTO DATASET

Class	S^2ENet	GLT	MFT	CALC	Exvit	FDNet	CrossHL	S3F3Net	MCFNet	Proposed
1	99.06	98.60	95.74	**99.74**	99.40	99.58	99.80	99.13	99.40	99.48
2	97.39	98.72	98.96	98.33	98.75	98.96	96.57	99.10	99.34	**99.41**
3	**100.00**	99.89	97.82	98.26	98.37	99.56	99.56	99.35	99.78	**100.00**
4	99.96	**100.00**	99.96	**100.00**	**100.00**	99.90	99.98	**100.00**	**100.00**	**100.00**
5	99.99	**100.00**	**100.00**	99.99	99.53	98.92	99.92	**100.00**	**100.00**	**100.00**
6	95.56	95.53	92.77	94.22	96.13	92.23	90.11	91.03	93.88	96.10
OA(%)	99.14	99.28	98.53	99.17	99.20	98.62	98.57	98.85	99.22	**99.47**
AA(%)	98.66	98.79	97.54	98.43	98.69	98.19	97.66	98.10	98.73	**99.16**
$\kappa\times100$	98.86	99.04	98.04	98.90	98.94	98.16	98.09	98.46	98.95	**99.29**

the land classification tasks with significant spatial variation features.

2) t-SNE Visualization Analysis: To visually demonstrate the feature representation ability of the proposed method, Fig. 7 shows the t-SNE dimensionality reduction visualization results of the original features and three advanced methods (including the proposed ACFFN) on the Trento dataset. It can be seen that the original features exhibit obvious intraclass divergence and interclass overlap, and the boundaries of each category are blurred. After being processed by various methods, the feature distribution shows an optimization trend of intraclass aggregation and interclass separation. Among them, although MCFNet can expand the interclass distance, the intraclass distance is relatively large, and GLT shows good intraclass compactness through multiscale feature modeling. In contrast, the proposed ACFFN exhibits superior feature clustering performance: clear interclass boundaries, highly aggregated intraclass samples into a single cluster structure, and successfully separated the feature overlap between the third and sixth classes. The above results indicate that ACFFN can effectively explore the inherent complementarity of multimodal data and enhance the representation ability of features.

E. Analysis of Complexity

To conduct an objective assessment of the computational efficiency of the proposed method, we compared the complexity of each model on the Houston 2013 dataset using

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TABLE V
CLASSIFICATION RESULTS OF MUUFL DATASET

Class	S^2ENet	GLT	MFT	CALC	Exvit	FDNet	CrossHL	S3F3Net	MCFNet	Proposed
1	87.74	89.94	89.56	84.47	90.34	85.84	**90.49**	86.75	88.47	88.22
2	74.64	76.45	71.73	79.81	60.28	76.14	74.75	76.24	**84.09**	70.16
3	64.59	65.75	60.33	73.10	70.63	66.50	71.97	63.96	65.13	**75.68**
4	93.58	91.10	92.12	69.93	93.41	80.51	82.26	91.58	95.13	**95.40**
5	86.35	85.89	75.03	84.13	83.45	82.84	85.53	86.89	83.62	**87.70**
6	**100.00**	**100.00**	99.93	**100.00**	98.73	**100.00**	97.98	99.78	**100.00**	99.78
7	91.05	91.10	87.11	94.44	91.13	**95.26**	88.63	91.41	94.17	93.09
8	92.04	92.55	90.86	94.39	93.74	92.89	94.22	**95.35**	94.28	94.29
9	69.30	56.41	53.21	68.13	57.02	66.20	56.46	70.62	55.97	**72.60**
10	85.28	56.03	74.23	87.12	64.83	87.12	68.51	87.73	60.12	**90.80**
11	98.80	97.86	98.80	98.80	98.80	97.59	97.86	96.39	**99.20**	98.39
OA(%)	84.06	84.81	81.87	83.46	84.27	83.06	85.48	84.11	85.14	**86.01**
AA(%)	85.76	82.10	81.17	84.93	82.03	84.63	82.60	86.06	83.65	**87.83**
$\kappa\times100$	79.47	80.40	76.69	78.92	79.66	78.36	81.19	79.63	80.89	**81.97**

TABLE VI
CLASSIFICATION RESULTS OF SZU DATASET

Class	S^2ENet	GLT	MFT	CALC	Exvit	FDNet	CrossHL	S3F3Net	MCFNet	Proposed
1	94.38	92.88	90.81	93.47	94.41	92.87	92.20	89.64	**94.76**	93.17
2	88.68	88.14	88.68	85.35	89.84	**90.08**	88.76	86.12	87.13	89.77
3	87.79	88.01	90.70	91.48	89.28	90.06	89.28	**92.41**	90.49	90.28
4	82.77	80.78	84.68	84.16	80.18	87.21	77.79	84.12	77.33	**90.15**
5	91.70	90.92	89.70	92.13	**94.42**	84.62	90.06	94.28	88.13	92.42
6	69.70	**75.76**	72.73	74.24	73.48	65.15	72.73	74.24	72.73	74.24
7	85.03	78.04	78.85	80.81	79.19	83.64	79.12	83.14	**87.96**	84.88
8	94.92	95.22	95.36	94.85	96.03	94.85	**96.98**	94.78	95.36	96.32
9	92.18	91.08	93.94	**96.93**	93.26	95.34	88.82	96.77	91.15	93.85
10	93.42	91.10	90.45	**95.23**	94.45	93.03	94.71	92.77	93.68	94.58
11	92.70	91.30	91.52	91.47	89.53	**95.44**	92.97	93.93	93.08	93.67
12	96.14	**96.43**	94.28	91.05	94.77	95.60	95.18	93.98	95.17	95.18
13	86.07	82.26	83.96	80.92	87.85	88.29	80.22	80.10	86.04	**91.13**
14	94.74	91.91	91.08	93.38	89.61	95.34	95.32	**95.51**	93.55	94.90
15	98.32	97.90	97.90	98.32	97.48	97.48	97.06	98.32	96.64	**98.74**
16	96.44	92.65	96.21	**97.55**	96.88	97.10	97.33	97.10	97.33	97.10
17	96.75	95.08	93.27	95.30	**97.97**	97.38	92.73	96.43	91.33	96.75
18	86.73	90.76	93.48	77.35	89.39	92.95	90.13	94.37	87.51	**94.94**
19	98.82	99.64	98.58	99.88	**100.00**	99.64	99.76	99.88	98.93	**100.00**
20	98.31	96.91	97.69	98.18	96.85	98.77	95.60	**99.43**	97.27	99.30
OA(%)	89.09	87.03	88.33	87.72	87.68	90.88	86.15	88.96	87.53	**92.24**
AA(%)	91.28	90.34	90.69	90.60	91.24	91.74	90.34	91.87	90.78	**93.07**
$\kappa\times100$	87.69	85.41	86.84	86.11	86.12	89.65	84.44	87.54	85.99	**91.20**

parameters, floating-point operations, and training time per epoch as indicators. As shown in Table VII, S2ENet and MFT have the lowest computational cost, but their lightweight architecture limits their ability to effectively capture complex features, ultimately resulting in limited classification performance. GLT and S3F2net require the longest running time due to their multiscale feature processing and decision-level fusion mechanisms, respectively. The proposed method has a high

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TABLE VII
COMPARISON OF FLOPS, PARAMETERS, AND TRAINING TIME ON THE HOUSTON 2013 DATASET

Complexity	S^2ENet	GLT	MFT	CALC	EXvit	FDNet	CrossHL	S3F2Net	MCFNet	Proposed
FLOPs(M)	27.08	156.54	21.06	28.75	27.13	35.76	41.90	57.50	39.06	86.04
Parameters(M)	270.85	560.77	313.07	284.14	229.10	345.86	432.75	277.59	252.31	572.37
Training time(s)	0.26	1.62	0.25	0.31	0.52	0.37	0.28	1.74	0.37	0.41

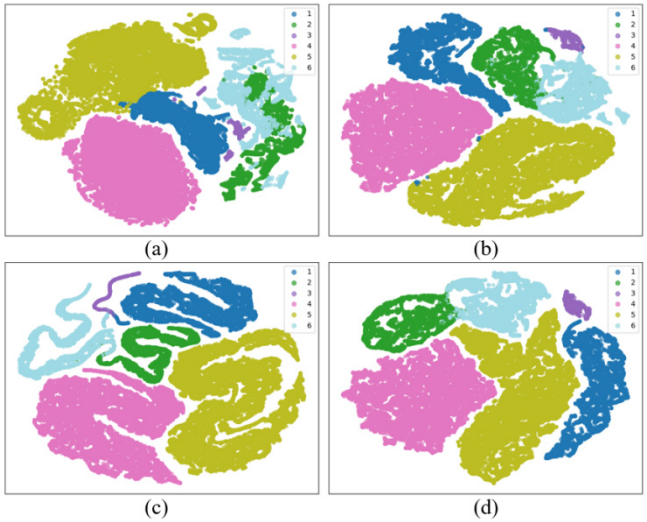


Fig. 7. t-SNE visualization results on the Trento dataset. (a) Original features. (b) GLT. (c) MCFNet. (d) Proposed.

number of parameters due to the introduction of the contourlet transform, but due to a carefully designed dynamic collaborative mechanism, it maintains reasonable computational costs while ensuring excellent classification performance.

V. CONCLUSION

This article proposes a new ACFFN for joint classification of hyperspectral and LiDAR data. To capture the inherent complex spatial features of HSI, an ASSC module is proposed, which combines LiDAR geometric priors to achieve collaborative fusion of spatial spectral features. To explore cross-modal complementary relationships, a feature fusion mechanism based on fuzzy learning is proposed, which introduces uncertainty distribution to achieve adaptive fusion of cross-modal features. In addition, an MGA module was designed to enhance the structural sensitivity of elevation information. Experiments on four publicly available datasets have shown that the ACFFN has excellent feature-learning capabilities compared to other advanced classification methods.

In future work, we will explore more robust cross-modal correlation modeling techniques, optimize the dynamic adaptation mechanism of multimodal feature fusion, and further improve the classification performance and training efficiency of the model.

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